Sophie (CSA)
Payments MFE
Payments BFF
IdentityServer
Payments API (PaymentSaga)
EwpPaymentsDb
Outbox publisher (in API)
Saga step runner (in API)
Kafka
Accounts Command Subscriber
Accounts API
Payments Saga Reply Subscriber
Payment Network Simulator
Phase 1 - Transfer pressed, the HTTP call ends in milliseconds
1 Review payment, press Transfer
2 POST /api/payments (cookie, X-CSRF, Idempotency-Key)
3 POST /v1/payments with SOPHIE's access token
4 Authorize (payment.initiate, branch) and validate
5 ONE transaction - payment INITIATED + saga RESERVE_FUNDS + outbox ReserveFunds
6 202 Accepted, paymentId 42
7 202 Accepted, paymentId 42
8 Open status page for payment 42
Phase 2 - Reserve funds (command to Accounts, reply back)
9 Read unsent outbox rows
10 ReserveFunds to accounts.commands
11 ReserveFunds
12 M2M token (client credentials, cached)
13 POST /internal/v1/funds/command-s
14 Inbox check, reserve 200, outbox FundsReserved
15 200 OK
16 Commit offset
Accounts' own outbox publisher sends FundsReserved
17 FundsReserved
18 M2M token (cached)
19 POST /internal/v1/payment-sagas/replies
20 Inbox check, load saga 42, DECIDE - no approval needed, save step SEND_TO_NETWORK
21 200 OK
22 Commit offset
Phase 3 - Send to the outside network (direct HTTP, no Kafka)
23 Pick due sagas (FOR UPDATE SKIP LOCKED)
24 POST /payments (Idempotency-Key, timeout, retry, circuit breaker)
25 Sent
26 DECIDE - save step SETTLE_FUNDS + outbox SettleFunds
Phase 4 - Settle funds (same round trip as phase 2)
27 SettleFunds to accounts.commands
28 SettleFunds
29 POST /internal/v1/funds/command-s
30 Debit the account, outbox FundsSettled
31 FundsSettled
32 POST /internal/v1/payment-sagas/replies
33 DECIDE - payment COMPLETED, saga COMPLETED, outbox PaymentCompleted
Phase 5 - Sophie sees the result live
34 PaymentCompleted to payments.payment.completed
Notifications subscriber, Notifications API, SignalR, Shell, BSS_NOTIFICATION to the MFE
35 GET /api/payments/42
36 GET /v1/payments/42 with Sophie's token
37 COMPLETED, with the saga timeline
38 Funds reserved, Sent, Settled, Completed
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